CM5 socket, M.2 connector

File: cm5_socket.kicad_sch

System MCU, Watchdog, EEPROM

File: system_mcu.kicad_sch

AC power, DC power, AC sync

File: ac_dc_power.kicad_sch

Isolated digital IO, Fan controller, Encoder

File: digital_io_fan_encoder.kicad_sch

7-port USB3 Hub

File: usb3_hub.kicad_sch

Rear USB, USB-C

File: usb_conn.kicad_sch

Ethernet, Video outputs, Ref clk

File: video_outputs.kicad_sch

DIB 2.0

File: dib.kicad_sch

Power modules coupling

File: power_coupling.kicad_sch

USB Port	USB type	STM32H7 CM5	Raspberry PI CM5
Rear top	USB A	FS only	USB #0, HS/SS
Rear bottom	USB A	FS/HS	USB Hub P1 Flex
Front	USB C	USB Hub P2, HS/SS, Dual role	
DIB #1	DIB 2.0	USB Hub P7, HS/SS	
DIB #2	DIB 2.0	USB Hub P6, HS/SS	
DIB #3	DIB 2.0	USB Hub P5, HS/SS	
DIB #4	DIB 2.0	USB Hub P4, HS/SS	
DIB #5	DIB 2.0	USB Hub P3, HS/SS	

I2C address	Device
0x51	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x7A	PTN5150A USB CC controller

H1 H2 H3 H4 H5 H6 H7 H8

PWR_FLAG

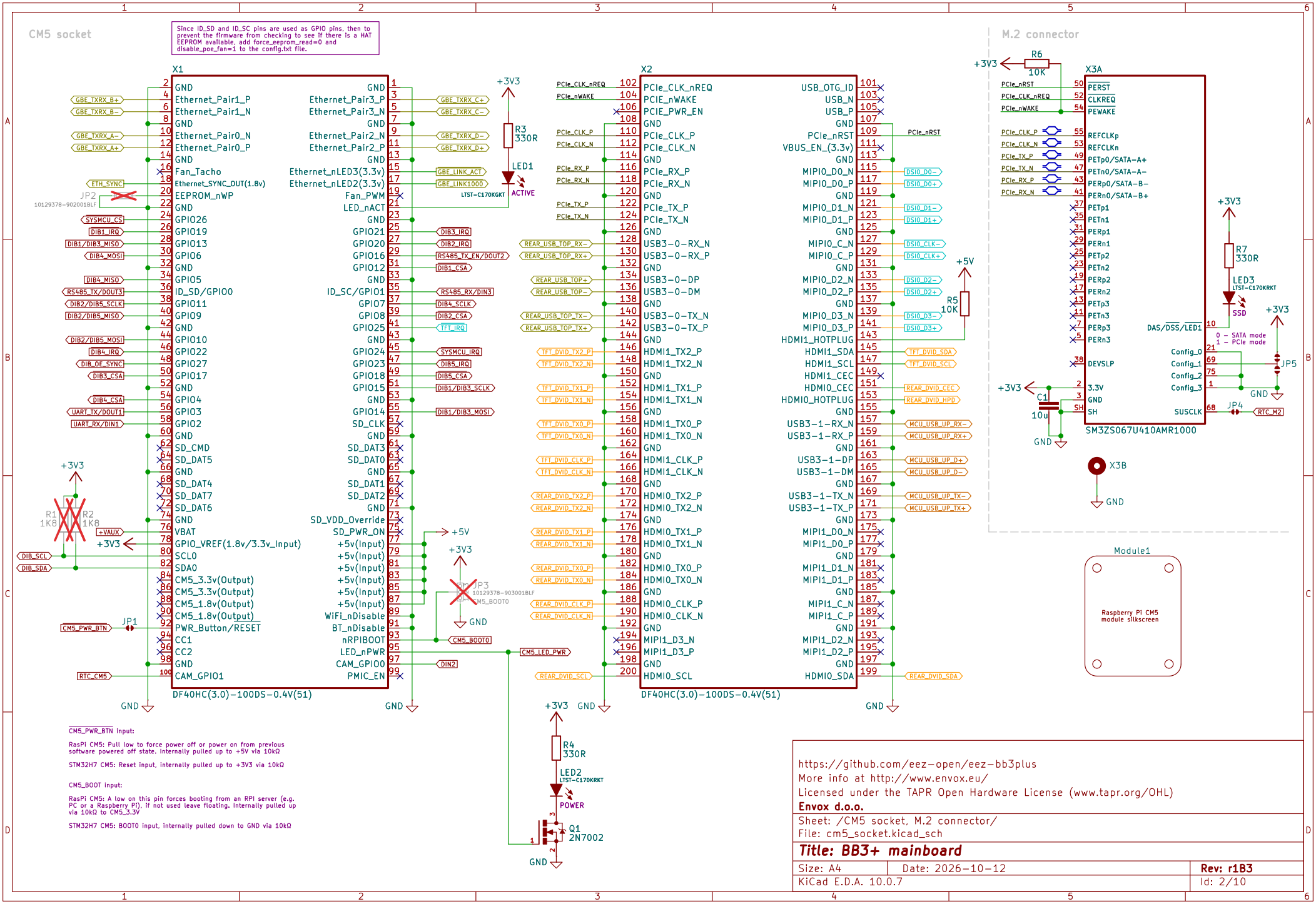
PE

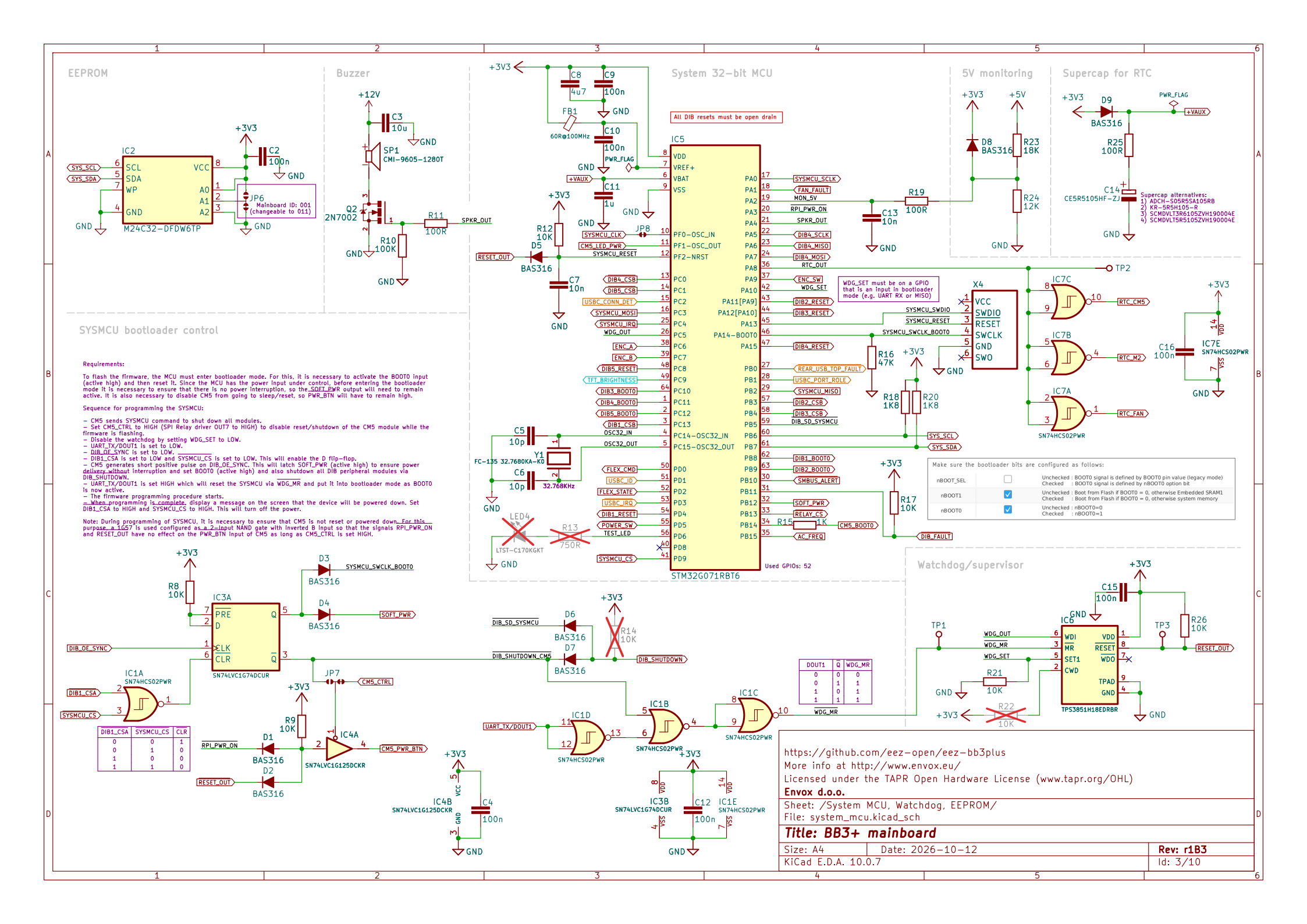
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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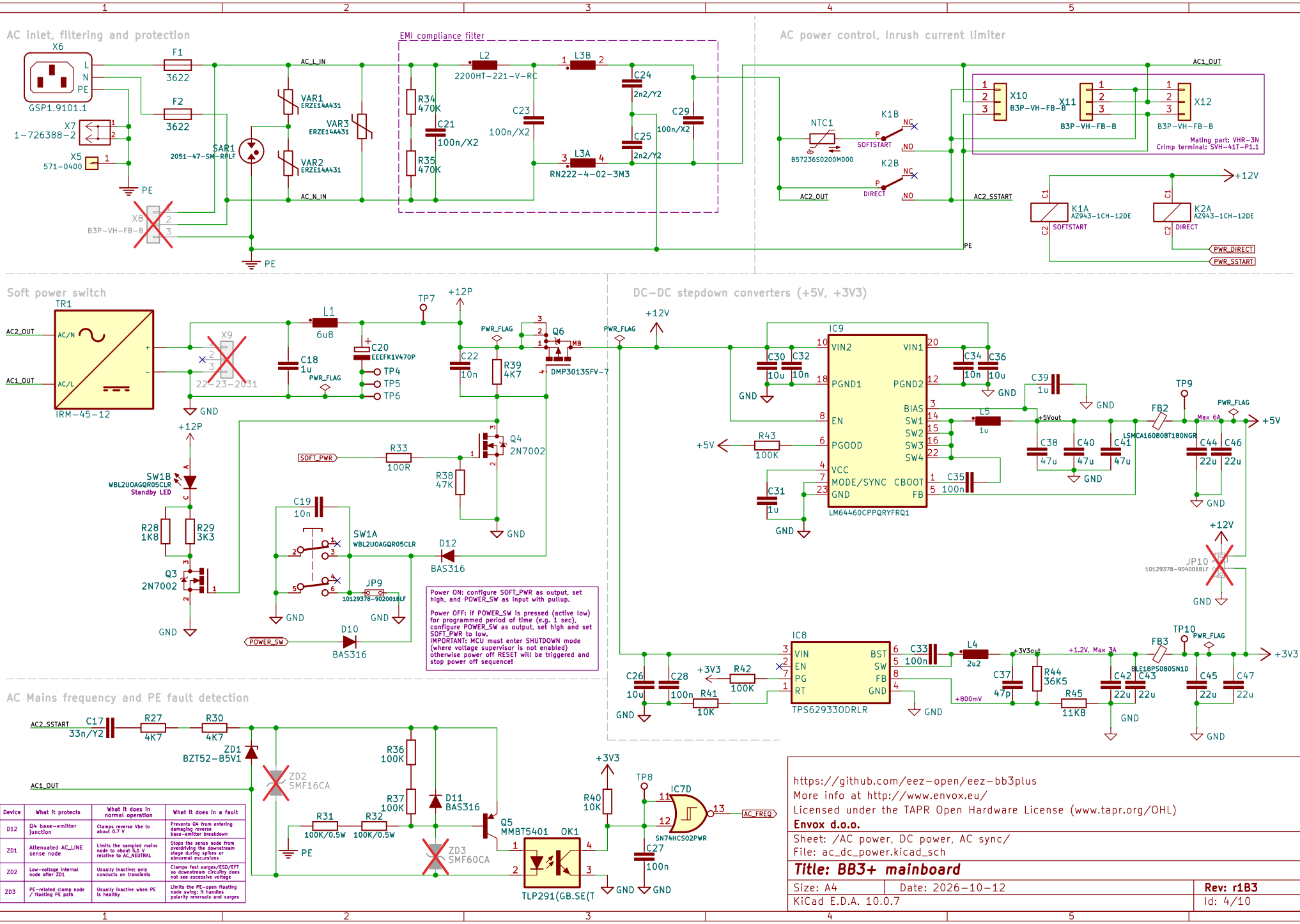
Sheet: /
File: BB3plus mainboard.kicad_sch

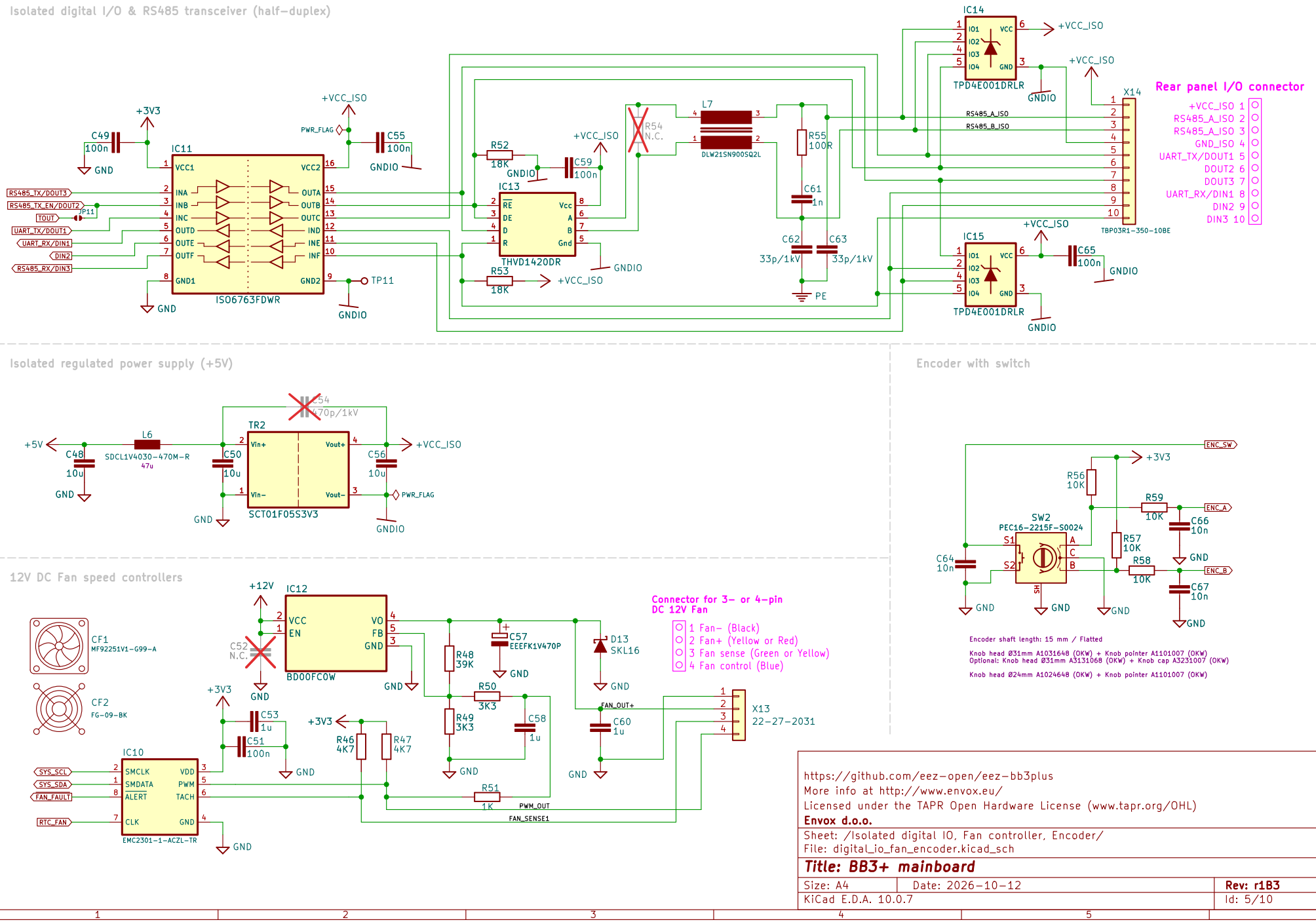
Title: BB3+ mainboard

Size: A4	Date: 2026-10-12	**Rev: r1B3**
KiCad E.D.A. 10.0.7		Id: 1/10


CM5 socket
Since ID_SD and ID_SC pins are used as GPIO pins, then to prevent the firmware from checking to see if there is a HAT EEPROM available, add force_eeprom_read=0 and disable_poe_fan=1 to the config.txt file.
X1
DF40HC(3.0)-100DS-0.4V(51)
X2
DF40HC(3.0)-100DS-0.4V(51)
M.2 connector
X3A
SM3ZS067U410AMR1000
X3B
Module1
Raspberry PI CM5 module silkscreen
R1 1K8
R2 1K8
R3 330R
R4 330R
R5 10K
R6 10K
R7 330R
C1 10u
LED1 ACTIVE
LED2 POWER
LED3 SSD
Q1 2N7002
JP1
JP2
JP3
JP4
JP5
CM5_PWR_BTN input:
RasPI CM5: Pull low to force power off or power on from previous software powered off state. Internally pulled up to +5V via 10kΩ
STM32H7 CM5: Reset input, internally pulled up to +3V3 via 10kΩ
CM5_BOOT input:
RasPi CM5: A low on this pin forces booting from an RPI server (e.g. PC or a Raspberry PI). If not used leave floating. Internally pulled up via 10kΩ to CM5_3.3V
STM32H7 CM5: BOOT0 input, internally pulled down to GND via 10kΩ
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Sheet: /CM5 socket, M.2 connector/
File: cm5_socket.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-12
Rev: r1B3
KiCad E.D.A. 10.0.7
Id: 2/10

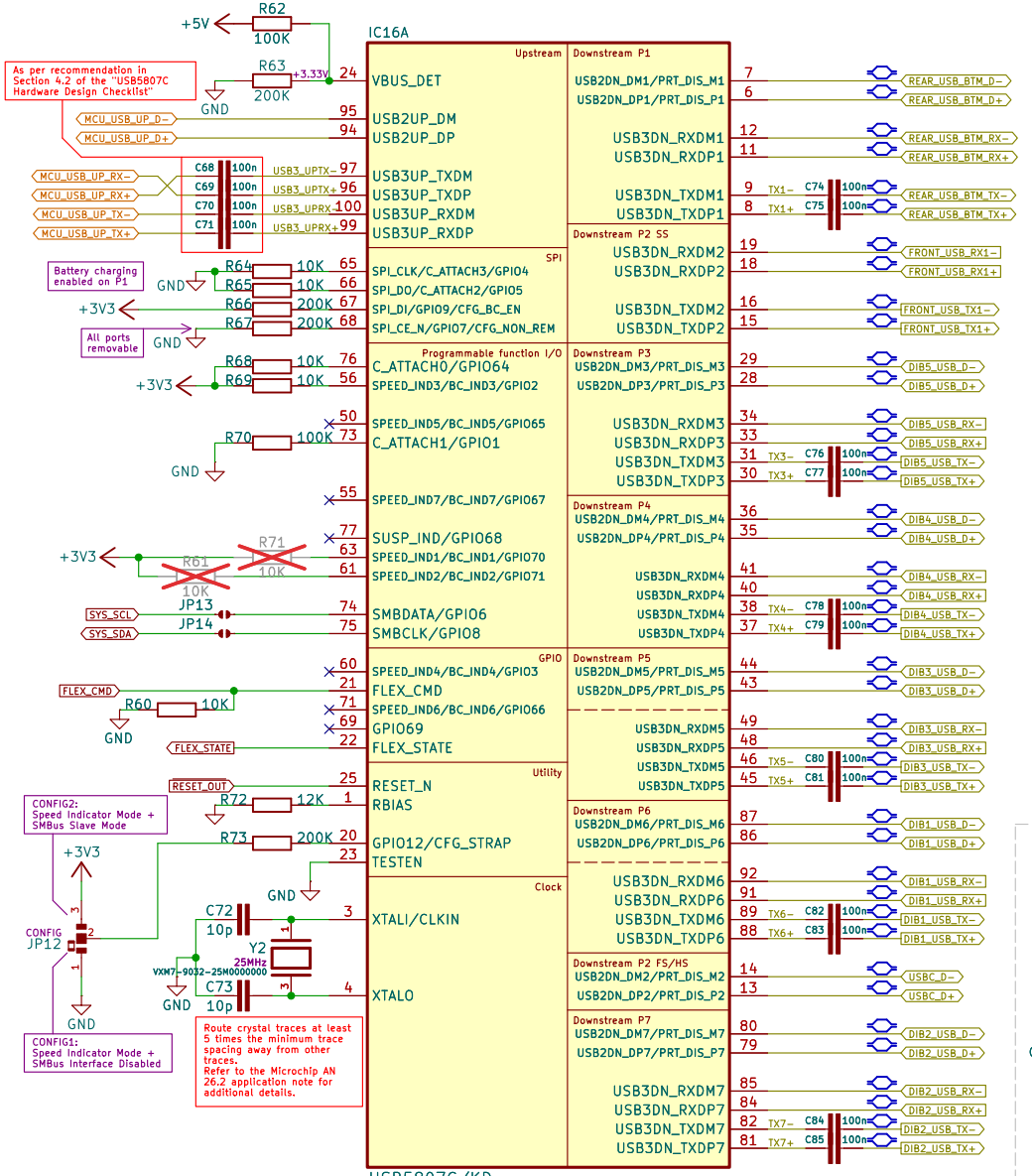

EEPROM
IC2
M24C32-DFDW6TP
JP6
Mainboard ID: 001 (changeable to 011)
Buzzer
SP1
CMI-9605-1280T
Q2
2N7002
System 32-bit MCU
IC5
STM32G071RBT6
All DIB resets must be open drain
Used GPIOs: 52
5V monitoring
Supercap for RTC
D9
BAS316
Supercap alternatives:
1) ADCH-SO5R5SA105RB
2) KR-5R5H105-R
3) SCMDVLT3R6105ZVH190004E
4) SCMDVLT5R5105ZVH190004E
SYSMCU bootloader control
Requirements:
To flash the firmware, the MCU must enter bootloader mode. For this, it is necessary to activate the BOOT0 input (active high) and then reset it. Since the MCU has the power input under control, before entering the bootloader mode it is necessary to ensure that there is no power interruption, so the SOFT_PWR output will need to remain active. It is also necessary to disable CM5 from going to sleep/reset, so PWR_BTN will have to remain high.
Sequence for programming the SYSMCU:
– CM5 sends SYSMCU command to shut down all modules.
– Set CM5_CTRL to HIGH (SPI Relay driver OUT7 to HIGH) to disable reset/shutdown of the CM5 module while the firmware is flashing.
– Disable the watchdog by setting WDG_SET to LOW.
– UART_TX/DOUT1 is set to LOW.
– DIB_OE_SYNC is set to LOW.
– DIB1_CSA is set to LOW and SYSMCU_CS is set to LOW. This will enable the D flip-flop.
– CM5 generates short positive pulse on DIB_OE_SYNC. This will latch SOFT_PWR (active high) to ensure power delivery without interruption and set BOOT0 (active high) and also shutdown all DIB peripheral modules via DIB_SHUTDOWN.
– UART_TX/DOUT1 is set HIGH which will reset the SYSMCU via WDG_MR and put it into bootloader mode as BOOT0 is now active.
– The firmware programming procedure starts.
– When programming is complete, display a message on the screen that the device will be powered down. Set DIB1_CSA to HIGH and SYSMCU_CS to HIGH. This will turn off the power.
Note: During programming of SYSMCU, it is necessary to ensure that CM5 is not reset or powered down. For this purpose, a 1G57 is used configured as a 2-input NAND gate with inverted B input so that the signals RPI_PWR_ON and RESET_OUT have no effect on the PWR_BTN input of CM5 as long as CM5_CTRL is set HIGH.
WDG_SET must be on a GPIO that is an input in bootloader mode (e.g. UART RX or MISO)
Make sure the bootloader bits are configured as follows:
nBOOT_SEL – Unchecked : BOOT0 signal is defined by BOOT0 pin value (legacy mode) Checked : BOOT0 signal is defined by nBOOT0 option bit
nBOOT1 – Unchecked : Boot from Flash if BOOT0 = 0, otherwise Embedded SRAM1 Checked : Boot from Flash if BOOT0 = 0, otherwise system memory
nBOOT0 – Unchecked : nBOOT0=0 Checked : nBOOT0=1
Watchdog/supervisor
IC6
TPS3851H18EDRBR
IC1A SN74HCS02PWR
IC1B SN74HCS02PWR
IC1C SN74HCS02PWR
IC1D SN74HCS02PWR
IC1E SN74HCS02PWR
IC3A SN74LVC1G74DCUR
IC3B SN74LVC1G74DCUR
IC4A SN74LVC1G125DCKR
IC4B SN74LVC1G125DCKR
IC7A SN74HCS02PWR
IC7B
IC7C
IC7E SN74HCS02PWR
X4
SWD
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Sheet: /System MCU, Watchdog, EEPROM/
File: system_mcu.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-12
Rev: r1B
KiCad E.D.A. 10.0.7
Id: 3/10


AC inlet, filtering and protection
EMI compliance filter
AC power control, Inrush current limiter
Soft power switch
DC–DC stepdown converters (+5V, +3V3)
AC Mains frequency and PE fault detection
X6
GSP1.9101.1
F1
3622
F2
3622
X7
1–726388–2
X5
571–0400
SAR1
2051–47–SM–RPLF
VAR1
ERZE14A431
VAR3
ERZE14A431
VAR2
ERZE14A431
X8
B3P–VH–FB–B
AC_L_IN
AC_N_IN
PE
L2
2200HT–221–V–RC
L3B
L3A
RN222–4–02–3M3
R34
470K
R35
470K
C21
100n/X2
C23
100n/X2
C24
2n2/Y2
C25
2n2/Y2
C29
100n/X2
NTC1
B57236S0200M000
K1B
K2B
SOFTSTART
DIRECT
X10
B3P–VH–FB–B
X11
B3P–VH–FB–B
X12
B3P–VH–FB–B
Mating part: VHR–3N
Crimp terminal: SVH–41T–P1.1
AC1_OUT
AC2_OUT
AC2_SSTART
K1A
AZ943–1CH–12DE
K2A
AZ943–1CH–12DE
+12V
PWR_DIRECT
PWR_SSTART
TR1
IRM–45–12
AC2_OUT
AC1_OUT
X9
22–23–2031
L1
6u8
C18
1u
C20
EEEFK1V470P
TP4
TP5
TP6
TP7
+12P
C22
10n
R39
4K7
PWR_FLAG
Q6
DMP3013SFV–7
GND
+12P
SW1B
WBL2UOAGQR05CLR
Standby LED
R28
1K8
R29
3K3
Q3
2N7002
SOFT_PWR
R33
100R
Q4
2N7002
R38
47K
C19
10n
SW1A
WBL2UOAGQR05CLR
JP9
10129378–902001BLF
D12
BAS316
D10
BAS316
POWER_SW
Power ON: configure SOFT_PWR as output, set high, and POWER_SW as input with pullup.
Power OFF: if POWER_SW is pressed (active low) for programmed period of time (e.g. 1 sec), configure POWER_SW as output, set high and set SOFT_PWR to low.
IMPORTANT: MCU must enter SHUTDOWN mode (where voltage supervisor is not enabled) otherwise power off RESET will be triggered and stop power off sequence!
IC9
LM64460CPPQRYFRQ1
VIN2
VIN1
PGND1
PGND2
BIAS
EN
SW1
SW2
SW3
SW4
PGOOD
VCC
MODE/SYNC
CBOOT
GND
FB
C30
10u
C32
10n
C34
10n
C36
10u
C39
1u
C31
1u
C35
100n
R43
100K
L5
1u
+5Vout
FB2
LSMCA160808T180NGR
Max 6A
+5V
C38
47u
C40
47u
C41
47u
C44
22u
C46
22u
TP9
JP10
10129378–904001BLF
IC8
TPS629330DRLR
VIN
EN
PG
RT
BST
SW
FB
GND
C26
10u
C28
100n
R41
10K
+3V3
R42
100K
C33
100n
L4
2u2
+3V3out
+1.2V, Max 3A
+800mV
C37
47p
R44
36K5
R45
11K8
FB3
BLE18PS080SN1D
TP10
+3V3
C42
22u
C43
22u
C45
22u
C47
22u
AC2_SSTART
C17
33n/Y2
R27
4K7
R30
4K7
ZD1
BZT52–B5V1
ZD2
SMF16CA
AC1_OUT
R36
100K
R37
100K
R31
100K/0.5W
R32
100K/0.5W
D11
BAS316
ZD3
SMF60CA
Q5
MMBT5401
OK1
TLP291(GB.SE(T
R40
10K
TP8
IC7D
SN74HCS02PWR
AC_FREQ
C27
100n
Device
What it protects
What it does in normal operation
What it does in a fault
D12
Q4 base–emitter junction
Clamps reverse Vbe to about 0.7 V
Prevents Q4 from entering damaging reverse base–emitter breakdown
ZD1
Attenuated AC_LINE sense node
Limits the sampled mains node to about 5.1 V relative to AC_NEUTRAL
Stops the sense node from overdriving the downstream stage during spikes or abnormal excursions
ZD2
Low–voltage internal node after ZD1
Usually inactive; only conducts on transients
Clamps fast surges/ESD/EFT so downstream circuitry does not see excessive voltage
ZD3
PE–related clamp node / floating PE path
Usually inactive when PE is healthy
Limits the PE–open floating node swing; it handles polarity reversals and surges
https://github.com/eez–open/eez–bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /AC power, DC power, AC sync/
File: ac_dc_power.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026–10–12
Rev: r1B3
KiCad E.D.A. 10.0.7
Id: 4/10


Isolated digital I/O & RS485 transceiver (half-duplex)
IC11
ISO6763FDWR
IC13
THVD1420DR
IC14
TPD4E001DRLR
IC15
TPD4E001DRLR
L7
DLW21SN900SQ2L
X14
TBP03R1-350-10BE
Rear panel I/O connector
+VCC_ISO 1
RS485_A_ISO 2
RS485_A_ISO 3
GND_ISO 4
UART_TX/DOUT1 5
DOUT2 6
DOUT3 7
UART_RX/DIN1 8
DIN2 9
DIN3 10
Isolated regulated power supply (+5V)
TR2
SCT01F05S3V3
L6
SDCL1V4030-470M-R
Encoder with switch
SW2
PEC16-2215F-S0024
Encoder shaft length: 15 mm / Flatted
Knob head Ø31mm A1031648 (OKW) + Knob pointer A1101007 (OKW)
Optional: Knob head Ø31mm A3131068 (OKW) + Knob cap A3231007 (OKW)
Knob head Ø24mm A1024648 (OKW) + Knob pointer A1101007 (OKW)
12V DC Fan speed controllers
IC12
BD00FC0W
IC10
EMC2301-1-ACZL-TR
CF1
MF92251V1-G99-A
CF2
FG-09-BK
Connector for 3- or 4-pin DC 12V Fan
1 Fan- (Black)
2 Fan+ (Yellow or Red)
3 Fan sense (Green or Yellow)
4 Fan control (Blue)
X13
22-27-2031
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Sheet: /Isolated digital IO, Fan controller, Encoder/
File: digital_io_fan_encoder.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-12
Rev: r1B3
KiCad E.D.A. 10.0.7
Id: 5/10

7-port USB hub

All downstream ports that are not used should be disabled (use I2C to configure).

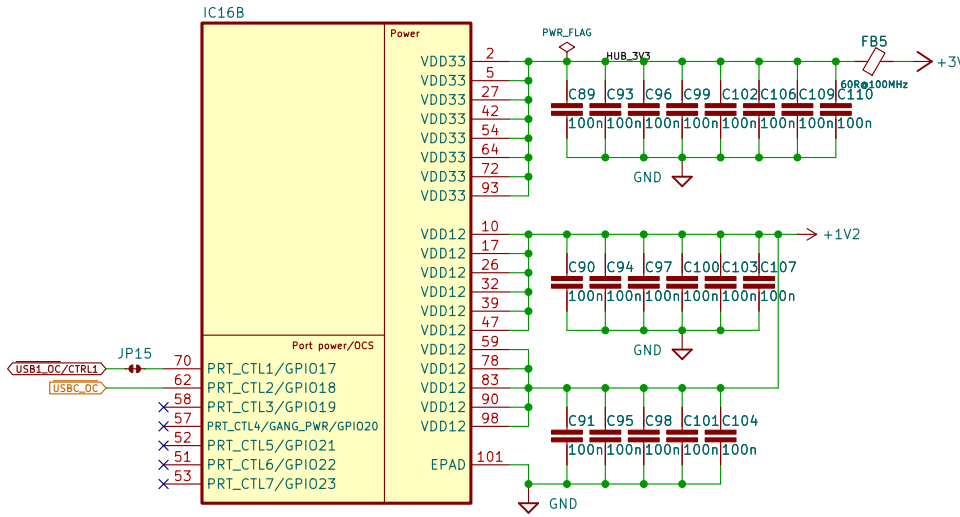


Microchip HUB troubleshoot articles:

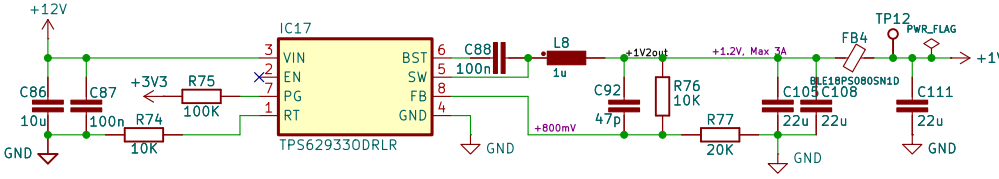
https://support.microchip.com/s/article/USB3-Hub-Upstream-Port-Connect-Enumeration-Failure
https://support.microchip.com/s/article/USB-High-Speed-Chirp-Explained
https://support.microchip.com/s/article/USB3-Connection-Process-and-Attach-Detection

IMPORTANT:

If I2C/SMBus pull-up resistors are detected by the USB5807C at start-up, the hub will wait indefinitely to be configured by the attached I2C/SMBus master. For early prototyping, it may be necessary to physically remove the pull-up resistors until the I2C/SMBus master is fully operational and able to properly configure the hub at start-up.



USB Hub DC-DC stepdown converter (+1V2)

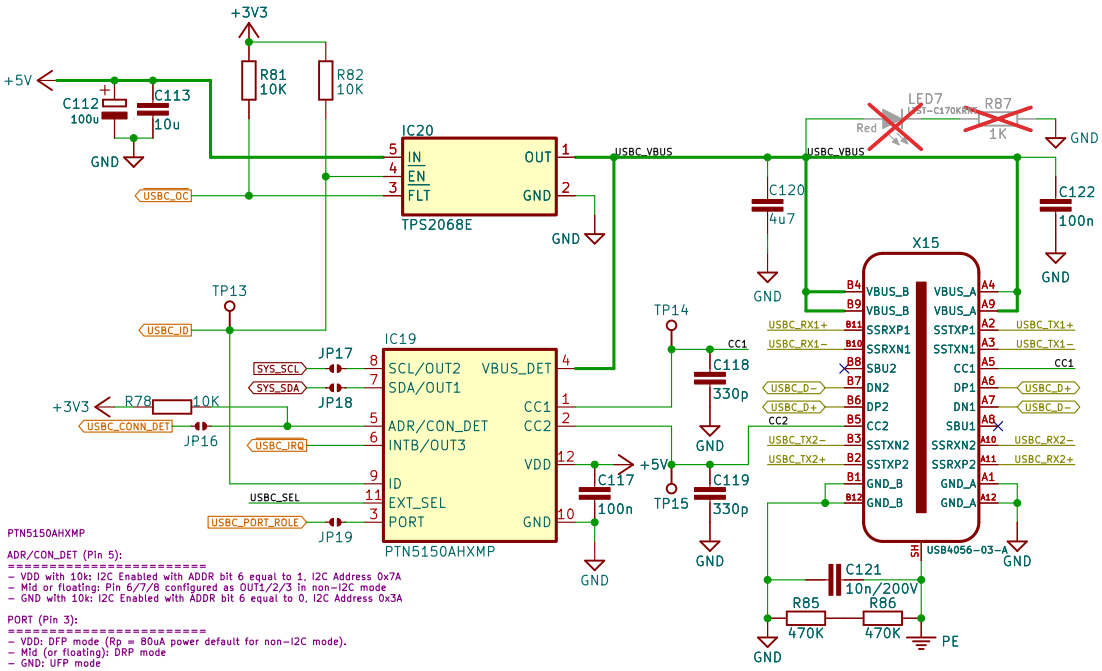


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Sheet: /7-port USB3 Hub/		
File: usb3_hub.kicad_sch		
Title: BB3+ mainboard		
Size: A4	Date: 2026-10-12	**Rev: r1B3**
KiCad E.D.A. 10.0.7		Id: 6/10

USB-C 3.0 front port (source max 1.5A)

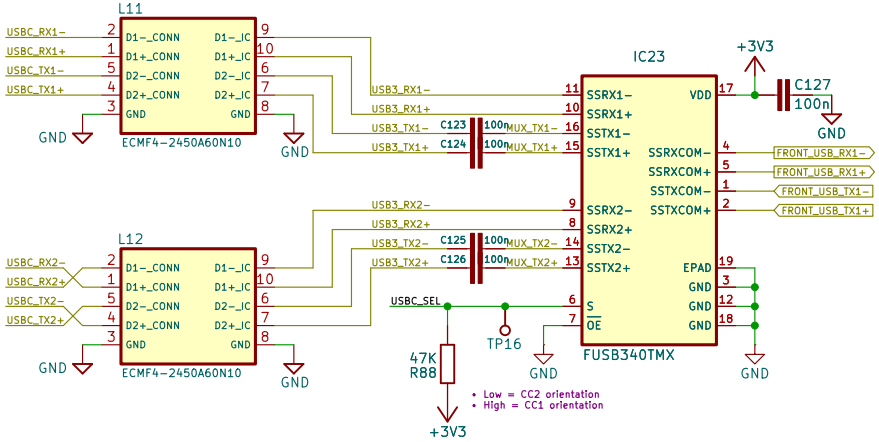


PTN5150AHXMP

ADR/CON_DET (Pin 5):
- VDD with 10k: I2C Enabled with ADDR bit 6 equal to 1, I2C Address 0x7A
- Mid or floating: Pin 6/7/8 configured as OUT1/2/3 in non-I2C mode
- GND with 10k: I2C Enabled with ADDR bit 6 equal to 0, I2C Address 0x3A

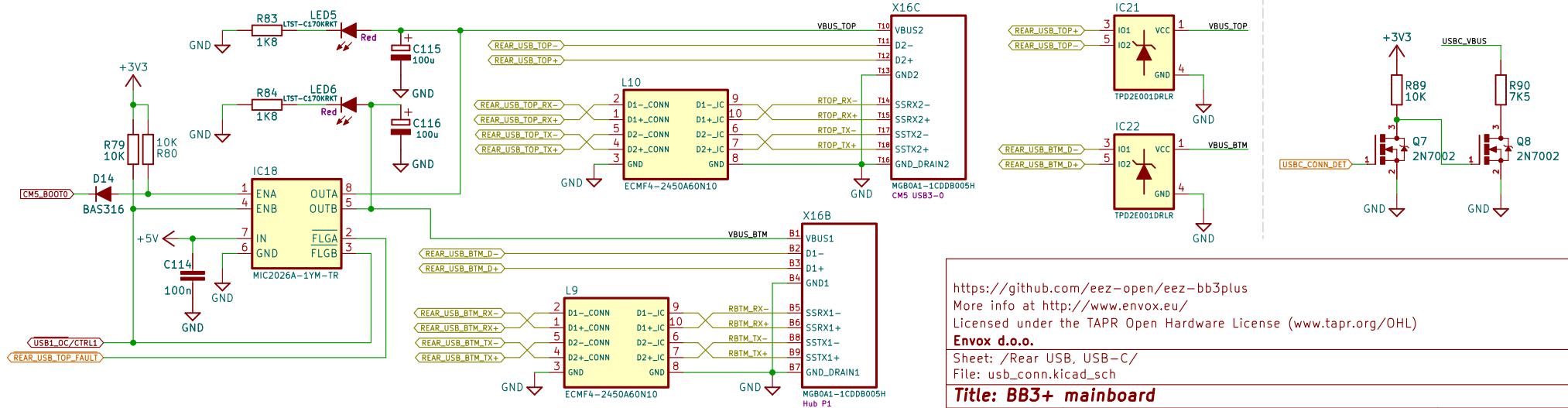
PORT (Pin 3):
- VDD: DFP mode (Rp = 80uA power default for non-I2C mode).
- Mid (or floating): DRP mode
- GND: UFP mode

USB-C 3.0 Protection and Mux



- Low = CC2 orientation
- High = CC1 orientation

USB2.0 Type-A (Rear panel)



VBus Discharge Circuit

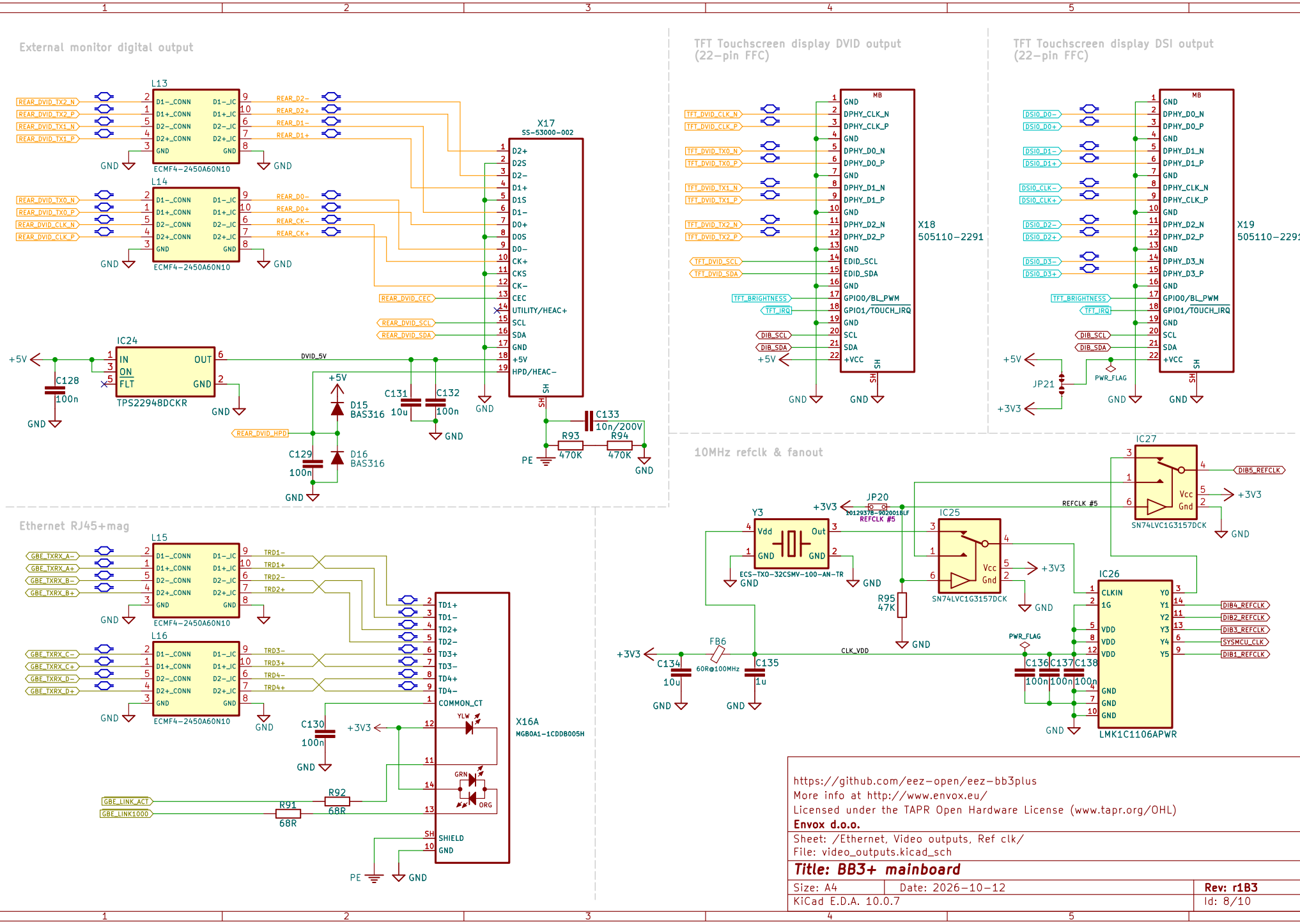
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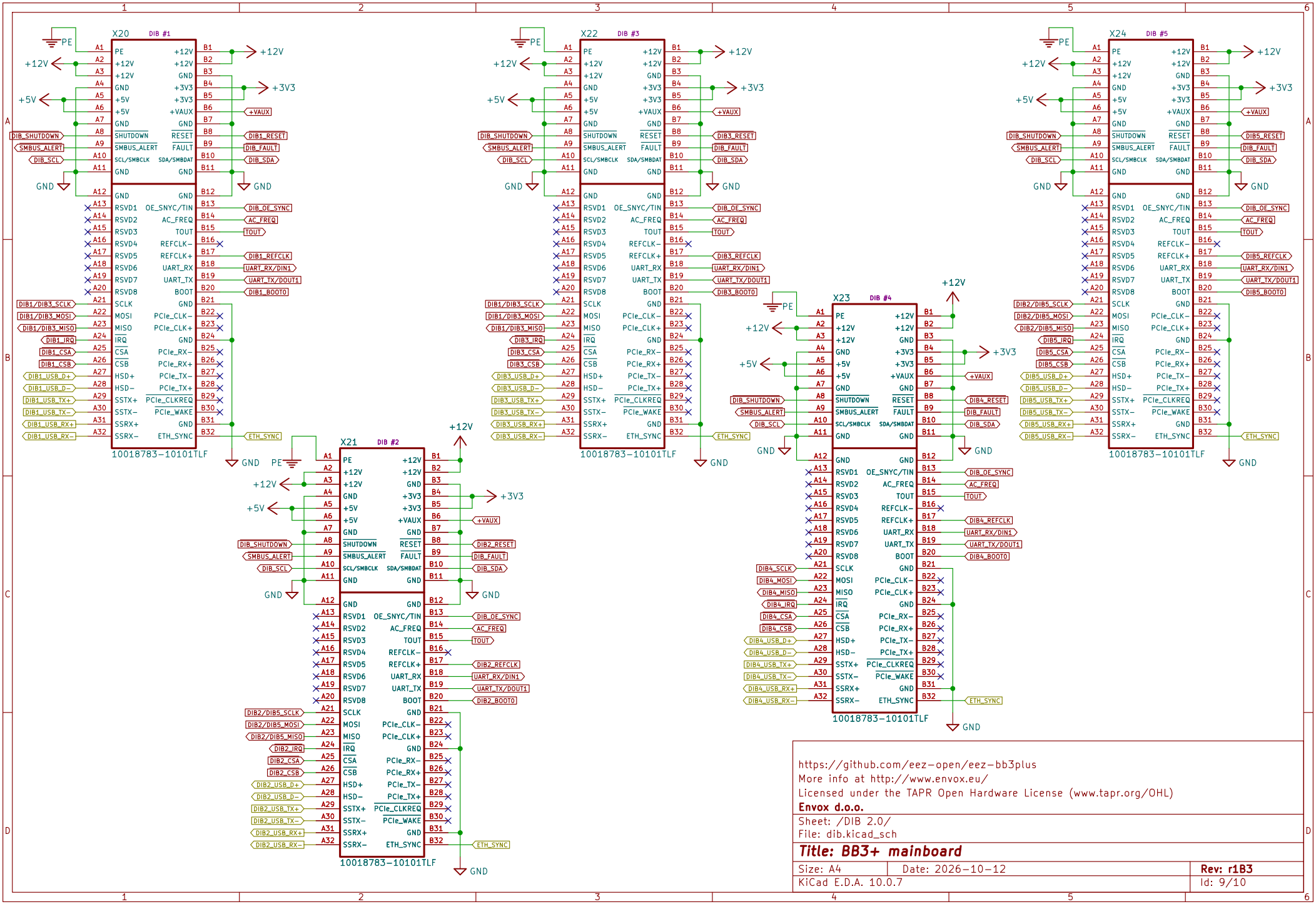
Envox d.o.o.

Sheet: /Rear USB, USB-C/
File: usb_conn.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-10-12	**Rev: r1B3**
KiCad E.D.A. 10.0.7		Id: 7/10

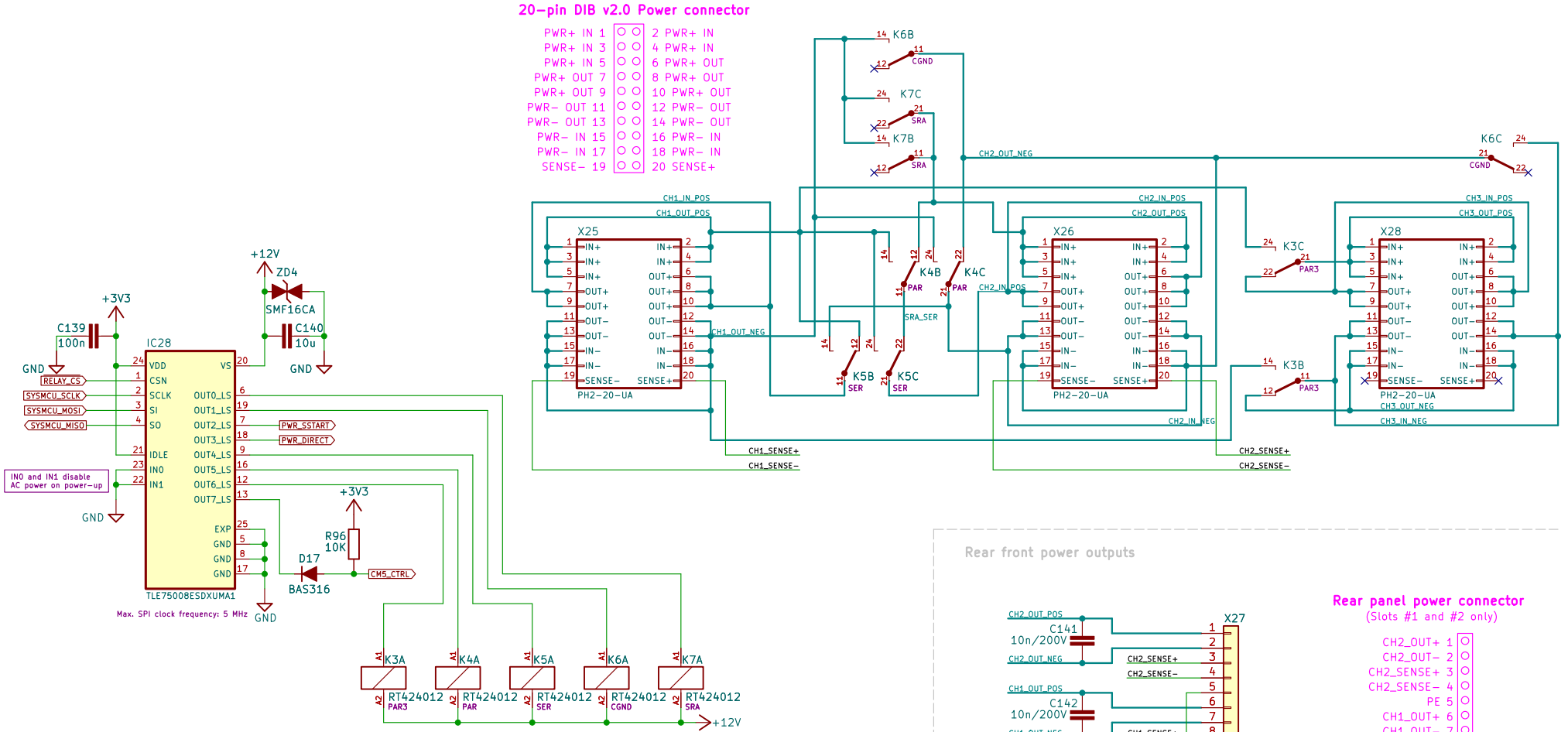

External monitor digital output
L13
L14
ECMF4-2450A60N10
X17
SS-53000-002
IC24
TPS22948DCKR
TFT Touchscreen display DVID output
(22-pin FFC)
X18
505110-2291
TFT Touchscreen display DSI output
(22-pin FFC)
X19
505110-2291
10MHz refclk & fanout
IC27
SN74LVC1G3157DCK
IC25
SN74LVC1G3157DCK
IC26
LMK1C1106APWR
Y3
ECS-TXO-32CSMV-100-AN-TR
JP20
JP21
FB6
Ethernet RJ45+mag
L15
L16
ECMF4-2450A60N10
X16A
MGB0A1-1CDDB005H
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /Ethernet, Video outputs, Ref clk/
File: video_outputs.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-12
Rev: r1B3
KiCad E.D.A. 10.0.7
Id: 8/10


X20
DIB #1
X21
DIB #2
X22
DIB #3
X23
DIB #4
X24
DIB #5
10018783-10101TLF
PE
+12V
+5V
+3V3
GND
+VAUX
ETH_SYNC
DIB_SHUTDOWN
SMBUS_ALERT
DIB_SCL
DIB_SDA
DIB_FAULT
DIB_OE_SYNC
AC_FREQ
TOUT
UART_RX/DIN1
UART_TX/DOUT1
DIB_SCLK
DIB_MOSI
DIB_MISO
DIB_IRQ
DIB_CSA
DIB_CSB
DIB_USB_D+
DIB_USB_D-
DIB_USB_TX+
DIB_USB_TX-
DIB_USB_RX+
DIB_USB_RX-
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Sheet: /DIB 2.0/
File: dib.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-12
Rev: r1B3
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Id: 9/10

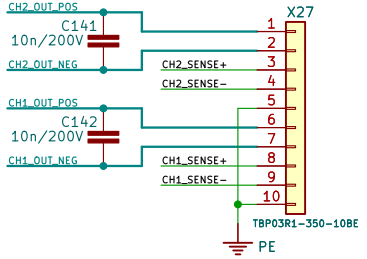
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	○○	2	PWR+ IN
PWR+ IN 3	○○	4	PWR+ IN
PWR+ IN 5	○○	6	PWR+ OUT
PWR+ OUT 7	○○	8	PWR+ OUT
PWR+ OUT 9	○○	10	PWR+ OUT
PWR- OUT 11	○○	12	PWR- OUT
PWR- OUT 13	○○	14	PWR- OUT
PWR- IN 15	○○	16	PWR- IN
PWR- IN 17	○○	18	PWR- IN
SENSE- 19	○○	20	SENSE+



Rear front power outputs



Rear panel power connector

(Slots #1 and #2 only)

CH2_OUT+	1
CH2_OUT-	2
CH2_SENSE+	3
CH2_SENSE-	4
PE	5
CH1_OUT+	6
CH1_OUT-	7
CH1_SENSE+	8
CH1_SENSE-	9
PE	10

https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.

Sheet: /Power modules coupling/
File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-10-12	**Rev: r1B3**
KiCad E.D.A. 10.0.7		Id: 10/10